

ECON 695 — Midterm 1 Formula Sheet (Lectures 1–4)

No vectors; core identities, workflows, and labels.

A. Setup & Notation

All sums are over $i = 1, \dots, N$. Sample means: $\bar{y} = \frac{1}{N} \sum y_i$, $\bar{x} = \frac{1}{N} \sum x_i$.

Sample variance/covariance:

$$\widehat{\text{Var}}(x) = \frac{1}{N} \sum (x_i - \bar{x})^2, \quad \widehat{\text{Cov}}(x, y) = \frac{1}{N} \sum (x_i - \bar{x})(y_i - \bar{y}).$$

Population orthogonality in $y = \beta_0 + \beta_1 x + u$: $\mathbb{E}[u] = 0$, $\mathbb{E}[ux] = 0$. *Linear projection*: $\mathbb{E}[y|x] = \beta_0 + \beta_1 x$ minimizes $\mathbb{E}[(y - \beta_0 - \beta_1 x)^2]$.

Demean identities: $\sum (x_i - \bar{x}) = 0$, $\sum (y_i - \bar{y}) = 0$. $\widehat{\text{Cov}}(x, y) = \frac{1}{N} \sum x_i y_i - \bar{x} \bar{y}$, $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2$.

B. Univariate OLS (with constant)

Model: $y_i = \beta_0 + \beta_1 x_i + u_i$; OLS minimizes $\frac{1}{N} \sum (y_i - b_0 - b_1 x_i)^2$.

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\widehat{\text{Cov}}(x, y)}{\widehat{\text{Var}}(x)}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

FOCs: $\sum \hat{u}_i = 0$, $\sum \hat{u}_i x_i = 0$, where $\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$. **Interpretation**: $\hat{\beta}_1$ = slope of best linear fit; $\hat{\beta}_0$ = fitted value at $\bar{x}$.

Example: if x increases by 1, $\hat{\beta}_1$ predicts average change in y . If $\widehat{\text{Cov}}(x, y) = 8$ and $\widehat{\text{Var}}(x) = 4$, then $\hat{\beta}_1 = 2$.

C. Univariate OLS (no constant)

$y_i = \beta x_i + u_i$ (no intercept).

$$\hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}, \quad \sum x_i (y_i - \hat{\beta} x_i) = 0.$$

Use this when all regressors are mean-zero or dummies spanning all groups.

D. Orthogonality Lemmas (Problem 3)

- **With constant**: residuals orthogonal to every regressor and to the constant: $\sum \hat{u}_i = 0$, $\sum \hat{u}_i x_{ji} = 0 \forall j$.
- **Adding orthogonal covariates**: If $\bar{x} = \bar{z} = 0$ and $\sum x_i z_i = 0$, then adding z leaves slope on x unchanged: $y = a_0 + a_1 x \Rightarrow \hat{a}_1 = \hat{b}_1$ in $y = b_0 + b_1 x + b_2 z$.
- **Dummy partitions**: If $D_g D_{g'} = 0$ ($g \neq g'$) and $\sum_g D_g = 1$, then each β_g measures group mean difference relative to the omitted group.

E. Frisch–Waugh–Lovell (FWL) Theorem

Coefficient on x_k in $y = \beta_0 + \beta_k x_k + \sum_{\ell \neq k} \beta_\ell x_\ell + u$:

1. Residualize x_k on $(1, x_{-k})$: $\hat{\xi}_i = x_{ki} - \widehat{\mathbb{E}}[x_{ki} | 1, x_{-k, i}]$.
2. Residualize y on $(1, x_{-k})$: $\hat{r}_i = y_i - \widehat{\mathbb{E}}[y_i | 1, x_{-k, i}]$.
3. Run $\hat{r}$ on $\hat{\xi}$ without a constant:

$$\hat{\beta}_k = \frac{\sum \hat{\xi}_i \hat{r}_i}{\sum \hat{\xi}_i^2} = \frac{\widehat{\text{Cov}}(y, \hat{\xi})}{\widehat{\text{Var}}(\hat{\xi})}.$$

FWL-twice workflow: (i) purge x_k , (ii) purge y , (iii) regress residuals.

Example: Target slope on x_1 in $y \sim (1, x_1, x_2)$: $x_1 = \hat{\pi}_0 + \hat{\pi}_2 x_2 + \hat{\xi}$, $y = \hat{\alpha}_0 + \hat{\alpha}_2 x_2 + \hat{r}$, then $\hat{\beta}_1 = \frac{\sum \hat{\xi}_i \hat{r}_i}{\sum \hat{\xi}_i^2}$.

F. Omitted-Variable Bias (short vs long)

Long: $y = \beta_0 + \beta_1 x + \beta_2 z + \varepsilon$, Short: $y = \lambda_0 + \lambda_1 x + u$, and $z = \pi_0 + \pi_1 x + \xi$ with $\sum \xi_i = \sum \xi_i x_i = 0$.

$$\lambda_1 = \beta_1 + \beta_2 \pi_1 \quad \text{sign bias} = \text{sign}(\beta_2) \times \text{sign}(\pi_1).$$

Example: $\beta_1 = 1.0$, $\beta_2 = 0.5$, $\pi_1 = 0.6 \Rightarrow \lambda_1 = 1.3$ (upward bias). If $\beta_2 < 0$ and $\pi_1 > 0$, $\lambda_1 < \beta_1$ (downward bias).

G. Oaxaca–Blinder (discrete X version)

For group a (private) and b (govt), cells $g = 1, \dots, G$. $p_g^a = \Pr(g|a)$, $p_g^b = \Pr(g|b)$, $\bar{y}_g^a = \mathbb{E}[y|g, a]$, $\bar{y}_g^b = \mathbb{E}[y|g, b]$.

Dummy OLS (no intercept): within group a , $y_i = \sum_g \bar{y}_g^a D_{gi} + u_i$.

Counterfactual: $\bar{y}_{cf}^b = \sum_g p_g^a \bar{y}_g^b = (\bar{x}^a)' \hat{\beta}^b$.

Gap decomposition:

$$\bar{y}^b - \bar{y}^a = \underbrace{(\bar{x}^b - \bar{x}^a)' \hat{\beta}^b}_{\text{composition}} + \underbrace{(\hat{\beta}^b - \hat{\beta}^a)' \bar{x}^a}_{\text{returns}}.$$

Composition = difference in mix; *Returns* = difference in pay within cells.

Example (3 cells):

g	1	2	3
p_g^a	0.3	0.4	0.3
p_g^b	0.2	0.5	0.3
$\bar{y}_g^a$	3.0	3.2	3.4
$\bar{y}_g^b$	3.1	3.3	3.6

$\bar{y}^a = 3.2$, $\bar{y}^b = 3.34$, gap = 0.14. Counterfactual: $\bar{y}_{cf}^b = \sum p_g^a \bar{y}_g^b = 0.3(3.1) + 0.4(3.3) + 0.3(3.6) = 3.33$. Composition = $\bar{y}^b - \bar{y}_{cf}^b = 0.01$, Returns = $\bar{y}_{cf}^b - \bar{y}^a = 0.13$.

Oaxaca workflow (exam quick):

1. Run group-specific dummy OLS (no intercept) $\Rightarrow \hat{\beta}_g^a, \hat{\beta}_g^b$.
2. Compute means $\bar{y}^a, \bar{y}^b$ and shares p_g^a, p_g^b .
3. Form counterfactual $\bar{y}_{cf}^b = \sum_g p_g^a \hat{\beta}_g^b$.
4. Report: composition, returns, total check.

H. Conceptual Quick Reference

- **Linear Projection (LP):** best linear predictor of y given x .
- **Orthogonality Condition:** $\mathbb{E}[xu] = 0 \Rightarrow$ unbiased OLS.
- **OLS Geometry:** $\hat{y} = X\hat{\beta}$ is projection of y onto $\text{span}(X)$; $\hat{u} \perp X$.
- **Dummy Variable Trap:** omit one category if constant included.
- **FWL Intuition:** partial effect = slope of residualized x_k - y regression.
- **OVB Intuition:** bias = effect of omitted variable $\times$ correlation with included regressor.
- **Oaxaca Labels:** “Between” = composition, “Within” = returns.
- **Exam Strategy:** If controls $\rightarrow$ use FWL; if variable omitted $\rightarrow$ apply OVB; if dummies $\rightarrow$ interpret as group means; for Oaxaca $\rightarrow$ cell means $\rightarrow$ counterfactual $\rightarrow$ composition/returns.

I. CEF with Dummies (Mutually Exclusive Groups)

Groups $G_0, G_1, \dots, G_J$ with dummies $D_{ji} = \mathbf{1}\{i \in G_j\}$. Model:

$$y_i = \beta_0 + \sum_{j=1}^J \beta_j D_{ji} + u_i.$$

Interpretation: $\beta_0 = \mathbb{E}[y|G_0]$, $\beta_j = \mathbb{E}[y|G_j] - \mathbb{E}[y|G_0]$. Each β_j is a mean gap vs. baseline group. If a control x_i is added:

$$y_i = \alpha_0 + \sum_{j=1}^J \alpha_j D_{ji} + \gamma x_i + \varepsilon_i,$$

then α_j measures the adjusted (partial) gap holding x_i fixed.

Example: y_i = wage, D_{1i} = immigrant; x_i = education. α_1 = conditional immigrant-native wage gap after controlling for education.

J. Oaxaca–Blinder Decomposition (continuous or discrete X)

Gap between two groups a, b :

$$\mathbb{E}[y|b] - \mathbb{E}[y|a] = \underbrace{(\bar{x}^b - \bar{x}^a)' \hat{\beta}^a}_{\text{composition}} + \underbrace{\bar{x}^b' (\hat{\beta}^b - \hat{\beta}^a)}_{\text{returns}}.$$

Choice of reference group (a or b) changes which part is interpreted as structure vs. endowment.

“Within/returns” = pay or effect differences, “Between/composition” = characteristic mix differences. $\beta_2 < 0$ and $\pi_1 > 0$, bias is downward.

Counterfactual mean: $\bar{y}_{cf}^b = (\bar{x}^a)' \hat{\beta}^b$.

Exam sanity check: composition + returns = total gap.

K. Typical Manipulations to Remember

- **De-meaning identity:** $\sum(x_i - \bar{x}) = \sum(y_i - \bar{y}) = 0$. Products expand to covariances.
- **Dummy algebra:** Mutually exclusive dummies satisfy $D_g D_{g'} = 0$ ($g \neq g'$), $\sum_g D_g = 1$.
- **Adding orthogonal covariates:** If $\bar{x} = \bar{z} = 0$ and $\frac{1}{N} \sum x_i z_i = 0$, adding z leaves slope on x unchanged.
- **FWL rule of thumb:** Whenever controls appear, residualize target regressor and (optionally) y , then run a one-regressor regression.

- **Demeaning trick (OLS with constant):** You can always remove intercept by centering variables: regress $\tilde{y}_i = (y_i - \bar{y})$ on $\tilde{x}_i = (x_i - \bar{x})$.

L. Worked Micro-Recipes (One-Liners)

- **FWL ratio:** From $x_k \sim (1, x_{-k})$ get residual $\hat{\xi}$ and report

$$\hat{\beta}_k = \frac{\sum \hat{\xi}_i y_i}{\sum \hat{\xi}_i^2} \quad (\text{no constant in final step}).$$

- **OVB sign test:** $\lambda_1 \geq \beta_1$ as $\beta_2 \pi_1 \geq 0$ where π_1 is slope in $z \sim (1, x)$.
- **Oaxaca short form:** Gap = $(\bar{x}^b - \bar{x}^a)' \hat{\beta}^b + (\hat{\beta}^b - \hat{\beta}^a)' \bar{x}^a$.
- **Demeaning check:** OLS with constant on (x, y) equals OLS without constant on demeaned data.
- **Two-step FWL numeric:** Residualize x_1 and y on x_2 : if $\text{Cov}(x_1, x_2) = 0$, step (ii) gives identical $\hat{\beta}_1$.

M. Frisch–Waugh Intuition (Geometry + Application)

OLS projects y onto the column space of X . FWL states that partialing out controls gives the same slope because projection residuals lie in orthogonal complement of the controls’ span.

$$\hat{\beta}_k = \frac{\langle M_{X_{-k}} y, M_{X_{-k}} x_k \rangle}{\langle M_{X_{-k}} x_k, M_{X_{-k}} x_k \rangle}, \quad M_{X_{-k}} = I - P_{X_{-k}}.$$

In scalar terms this reduces to residual-on-residual regression.

Exam note: if a question says “use FWL twice,” it means residualize both x_k and y on the same controls, then run $y \sim x_k$ with no constant.

N. Omitted Variable Bias Examples (Intuition Check)

1. If $\beta_2 > 0$ (omitted var has positive effect on y) and $\pi_1 > 0$ (correlated positively with x), then $\lambda_1 > \beta_1$ (upward bias).
2. If $\beta_2 < 0$ and $\pi_1 > 0$, bias is downward.
3. *Example:* True $y = 2 + 1.5x + 0.4z + u$, $z = 0.6 + 0.7x + \xi$ $\lambda_1 = 1.5 + 0.4(0.7) = 1.78$.

O. Linear vs. Conditional Expectation Function (CEF)

CEF: $\mathbb{E}[y|x]$ = population mean of y for each x . **Linear Projection (LP):** best linear approximation to CEF: $\mathbb{E}^*[y|x] = \beta_0 + \beta_1 x$ minimizes $\mathbb{E}[(y - \beta_0 - \beta_1 x)^2]$. OLS estimates LP, not necessarily true $\mathbb{E}[y|x]$.

Key link: If $\mathbb{E}[y|x]$ is linear, OLS is unbiased for the structural slope. If not, OLS gives the best linear predictor.

P. Exam Strategy Summary

- Controls $\rightarrow$ use FWL or residualize.
- Suspect missing variable $\rightarrow$ use OVB formula $\lambda_1 = \beta_1 + \beta_2\pi_1$.
- Dummies $\rightarrow$ interpret coefficients as mean gaps (adjusted if controls present).
- Oaxaca $\rightarrow$ compute group-specific dummy OLS, then counterfactual mean and split into composition/returns.
- When in doubt: check signs, write sample orthogonality, state FWL equivalence, and sanity-check totals.